

STOCK MARKET FUNDAMENTALS

1. Stock Market and Major Indian Exchanges

A stock market is a regulated platform where buyers and sellers trade shares of publicly listed companies. In India, the two primary exchanges are the National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE). For data analysis, these exchanges are the primary sources of structured market data, including historical OHLC (Open, High, Low, Close) prices, corporate action records, and trading volumes.

2. Major Market Indices

Indices like the Nifty 50 (NSE) and Sensex (BSE) track the performance of a specific basket of top companies (50 and 30, respectively) to represent overall market health. In data science, indices serve as benchmark features or target variables when building market-wide predictive models, or to normalize individual stock returns and remove systemic market noise.

3. Primary Market and IPOs

An Initial Public Offering (IPO) is the process where a private company offers shares to the public for the first time to raise capital. A SME IPO is a specialized SEBI-facilitated framework for Small and Medium Enterprises to list on dedicated exchange platforms, subject to different eligibility criteria than mainboard IPOs. In financial datasets, IPO dates are critical categorical features, as post-listing price behavior often exhibits distinct volatility patterns.

4. Basic Valuation and Company Metrics

These metrics are foundational features in any financial dataset:

- **Market Capitalization** – Total market value of outstanding shares. Formula: $\text{Current Share Price} \times \text{Total Outstanding Shares}$. Used to segment companies (Large, Mid, Small cap) for stratified sampling.
- **EPS (Earnings Per Share)** – Profit allocated to each share. Formula: $(\text{Net Income} - \text{Preferred Dividends}) / \text{Weighted Average Outstanding Shares}$. A core variable for profitability analysis.
- **P/E Ratio (Price-to-Earnings)** – Measures market willingness to pay per unit of earnings. Formula: $\text{Current Share Price} / \text{EPS}$. Useful for identifying relative valuation outliers in a sector.
- **P/B Ratio (Price-to-Book)** – Compares market value to book value. Formula: $\text{Current Share Price} / \text{Book Value Per Share}$. Often used in financial sector modeling where asset values are highly relevant.
- **Dividend** – A portion of profits distributed to shareholders. Dividend yield is a continuous feature indicating income generation.
- **Bonus Shares & Stock Split** – Corporate actions that increase share count without changing total market cap. Data pipelines must apply split adjustments to historical prices to prevent artificial drops in time-series data.
- **Trading Volume** – Total shares traded in a period. High volume often validates the statistical significance of a price movement in quantitative models.

5. Company Financial Statements

Financial statements are structured datasets released quarterly and annually:

- Balance Sheet – A snapshot of assets, liabilities, and equity at a specific point in time. It provides foundational data for leverage and liquidity features.
- Profit & Loss (P&L) Statement – Summarizes revenues, expenses, and profits over a period. It is the primary source for top-line (revenue) and bottom-line (net income) growth metrics.
- Cash Flow Statement – Tracks actual cash inflows and outflows across operating, investing, and financing activities. It is crucial for verifying if reported P&L profits are backed by real cash generation, helping to detect accounting anomalies.

6. Basic Financial Ratios

Ratios standardize financial statement line items, making them comparable across companies of different sizes, which is a necessary step for machine learning feature scaling. Common categories include:

- Profitability Ratios (e.g., Return on Equity, Net Profit Margin) – to assess earnings efficiency.
- Liquidity Ratios (e.g., Current Ratio) – to evaluate short-term debt-paying ability.
- Leverage Ratios (e.g., Debt-to-Equity) – to measure financial risk and capital structure.

7. Applied Example: Financial Statement Analysis (Infosys Ltd.)

When analyzing a large-cap company like Infosys Ltd., a data analyst extracts key variables from its official reports:

- P&L – Consistent revenue growth and stable net profit margins indicate operational efficiency.
- Balance Sheet – High cash reserves and minimal long-term debt reflect strong liquidity and low financial leverage.
- Cash Flow – Positive cash flow from operating activities confirms that reported profits are backed by actual cash generation, a key validation step in financial data cleaning.
- Ratios – A moderate historical P/E ratio and consistent dividend payouts provide baseline features for valuation models when compared to sector peers.

References

1. SEBI Investor Education: "Market Index – Sensex & Nifty" and "Basic Capital Structure & Financial Ratios".
2. National Stock Exchange (NSE) & Bombay Stock Exchange (BSE) official guidelines on Public Issues and SME IPO eligibility.
3. National Institute of Securities Markets (NISM): "Financial Statement Analysis: Meaning, Methods & Benefits".